

# JOEY LIU

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## Education

### Queen's University

**Anticipated Graduation: May 2027**

*Bachelor of Computer Science(Hons.)*

*GPA: 3.8/4.0*

- Data Structures and Algorithms, Discrete Structures, Computer Architecture, Database Management Systems, Statistics, Operating Systems, Software Architecture, Algorithms

## Experience

### Canadian Imperial Bank of Commerce (CIBC)

**September 2025 – December 2025**

*Data Analyst Intern*

*Toronto, Ontario*

- Designed a Python application automating employee offboarding emails, reducing manual communication overhead by 85%.
- Built an anomaly detection engine using deviation analysis across a 250k row CSV database to identify behavioral outliers.
- Automated ServiceNow whitelisting ticket workflow using Selenium and Python, saving 260+ hours of compliance review.

### CanDeal

**May 2025 – August 2025**

*Software Developer Intern*

*Toronto, Ontario*

- Developed a Go service consuming the Slack REST API to deliver server health alerts, enabling 24/7 on-call coverage.
- Connected a Python application to automate threat screening against AWS log data, saving 20+ hours of monthly review.
- Streamlined environment setup time by containerizing internal tools with Docker, standardizing deployments across the team.
- Offloaded 80% of configuration reconciliation across distributed servers by developing an internal Python validation tool.

## Projects

### ThelpoStreet | Python, Next.js, TypeScript, PostgreSQL, Flask, Selenium |

- Developed ThelpoStreet.com, a web platform that has ingested and displayed 4000+ SEC filings using OpenAI GPT-5 API.
- Wrote UI tests using Selenium WebDriver and pytest to validate frontend component rendering and user flows.
- Minimized redundant database API calls by 95% by implementing a Redis cache-aside pattern to avoid reprocessing filings.
- Cut infrastructure costs by 80% by offloading API handling to Cloudflare serverless functions for scalable deployment.

### Codey Village | Python, MongoDB, WebSockets, PyTest | | Google Developer Groups - Hackathon Track Winner

- Built a Python FastAPI backend to process Chrome extension activity events into a MongoDB-backed REST API.
- Engineered a WebSocket multiplayer system for up to 5 concurrent players, deployed on GCP with session affinity.
- Wrote a pytest integration test suite achieving 90% endpoint coverage across 60+ test cases covering happy paths and edge cases.

### BlockBlaster AI | C++, OpenCV, WebAssembly, React, Typescript, Vercel |

- Engineered a high-performance C++ application with a pipeline for screen capture, state extraction, and round-solving logic.
- Leveraged OpenCV.js image processing with machine learning brightness-variance classification to extract board state.
- Utilized heuristic-guided backtracking with branch pruning, reducing search space by 99.4% over a brute force method.

### FleetComm | C++, Go, React, ZeroMQ, Protobuf, CMake |

- Built a distributed traffic simulation with V2V communication and dynamic rerouting, improving fleet throughput by 22%.
- Designed a bidirectional Go gateway streaming simulation data over ZeroMQ PUB/SUB with browser command routing.
- Integrated Protocol Buffer deserialization to process and transform high-frequency C++ simulation frames at 30Hz for real-time downstream consumption.

## Technical Skills

- **Languages:** Python, Java, C++, GoLang, JavaScript, TypeScript, SQL, Bash
- **Technologies:** React, NodeJS, FastAPI, Flask, Tailwind, PostgreSQL, MySQL, MongoDB, Redis, Linux
- **Developer Tools:** Git, Jenkins, Jira, AWS, Docker, CI/CD, pytest, Selenium, Docker, NoSQL, Excel, CI/CD
- **Concepts:** REST/RPC APIs, Distributed Systems, Microservices, Software Testing, SDLC, Agile, DevOps, Automation